



Proposal No: 129954287B

**Metcalf Oil & Gas Co
MCKINNEY 1-10**

API # 21-066-22541-0000
KAWALSKI 11-27S-05W Field
11-27S-5W
Horicon County, Michigan
January 20, 2006

Acidizing Recommendation

Prepared for:
John Metcalf
Metcalf Oil & Gas

Prepared by:
JAMES T KIRK
Field Engineer
Sego Quadrant



Service Point:
Gaylord, Michigan
Bus Phone: (989) 732-7733
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Service Representatives:
Scotty McLaren
Account Manager
Fleet Command

Operator Name: Metcalf Oil & Gas Co
Well Name: MCKINNEY 1-10
Job Description: FOAMED ACID FRAC w/GEL DIVERTER (20K)
Date: January 20, 2006



Proposal No: 129954287B

JOB AT A GLANCE

Surface Treating Pressure (max)	5,450 psi	
Slurry HHP (avg)	584 hhp	
Total Rate (max)	11.70 bpm	
Slurry Rate (max)	8.00 bpm	
Nitrogen Rate (max)	14,912 scfm	
Estimated Pump Time (HH:MM)	00:50	
Nitrogen Pumped Volume	698,283 scf	
Nitrogen Cooldown Volume	60,000 scf	
Acid Frac Fluid	20,000 gals	28% HCL
Diverter	3,600 gals	X-linked Gel
Flush	1,260 gals	15# Gel

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WELL DATA

RESERVOIR DATA

Formation	Niagaran
Pay Zone Height	50 ft
MD Depth to Middle Perforation	7,381 ft
TVD Depth to Middle Perforation	6,616 ft
Reservoir Pressure	700 psi
Porosity	12 %
Fracture Gradient	1.05 psi/ft
Bottom Hole Fracture Pressure	6,947 psi
Bottom Hole Static Temperature	130 ° F

PERFORATED INTERVAL

DEPTH(ft)		Shots per Foot	Perf Diameter (in)	Total Perfs
MEASURED	TRUE VERTICAL			
7,030 - 7,031	6,623 - 6,623	1	0.38	1
7,130 - 7,131	6,621 - 6,621	1	0.38	1
7,230 - 7,231	6,619 - 6,619	1	0.38	1
7,330 - 7,331	6,617 - 6,617	1	0.38	1
7,430 - 7,431	6,615 - 6,615	1	0.38	1
7,530 - 7,531	6,613 - 6,613	1	0.38	1
7,630 - 7,631	6,611 - 6,611	1	0.38	1
7,730 - 7,731	6,609 - 6,609	1	0.38	1

Total Number of Perforations	8
Total Feet Perforated	8 ft

TUBULAR GEOMETRY

Tubing	2 3/8" O.D.	(1.995" I.D.)	4.7 #	<u>Top</u>	<u>Bottom</u>
				0	7,574
End of Tubing					7,574 ft
Pump Via					Tubing

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FLUID SPECIFICATIONS

Acid Frac Fluid: 28% HCL

Foam Volume: 20,000 Gallons
Pumped Liquid Volume: 7,000 Gallons
Pumped Gas Volume: 592 Mscf Nitrogen

Components:

100 %	HCL, 20.1 - 28%	Hydrochloric Acid
10 gpt	Ferrotrol 300L	Iron Control Product
4 gpt	FAW-5 (drum)	Foaming Agent
2.5 gpt	NE-940	Non-Emulsifier
2 gpt	CI-14	Corrosion Inhibitor

Diverter: X-linked Gel

Foam Volume: 3,600 Gallons
Pumped Liquid Volume: 1,260 Gallons
Pumped Gas Volume: 107 Mscf Nitrogen

Components:

3.75 gpt	GW-3LD Green	Gelling Agent
2.5 gpt	Enzyme G-NE	Gel Breaker
2.5 gpt	NE-940	Non-Emulsifier
1.5 gpt	XLW-32	Crosslinker
1.4 gpt	BF-7L	Buffers/Ph Control Product
1 gpt	Inflo-102	Surface Tension Reducer
0.05 gpt	Magnacide 575	Bacteria Control Product

Flush: 15# Gel

1,260 Gallons

Components:

3.75 gpt	GW-3LD Green	Gelling Agent
0.05 gpt	Magnacide 575	Bacteria Control Product

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ACID TREATMENT SCHEDULE

NITROGEN GAS SYSTEM

INPUT PARAMETERS

TVD Depth (Mid Perforation)	6,616 ft
MD Depth (Mid Perforation)	7,381 ft
Perforations Number	8
Perforation Diameter	0.375 in
Bottom Hole Frac Pressure	6,947 psi
Bottom Hole Static Temperature	130 ° F
Fluid Temperature in Tanks	50 ° F
Nitrogen Temperature at Surface	80 ° F

				<u>Top</u>	<u>Bottom</u>
Tubing	2 3/8" O.D.	(1.995" I.D.)	4.7 #	0	7,574

CALCULATED TEMPERATURES

	<u>Maximum</u>	<u>Minimum</u>
System at Wellhead	55 ° F	50 ° F
System at Perforation	57 ° F	57 ° F

CALCULATED RATES, PRESSURES & HHP REQUIREMENTS

	<u>Maximum</u>	<u>Minimum</u>	<u>Average</u>
Surface Treating Pressure (psi)	5,450	4,278	5,313
Fluid Rate (bpm)	8.0	4.2	4.5
Nitrogen Rate (scfm)	14,913	14,913	14,913
Fluid Hydraulic Horsepower	839	555	584
Nitrogen Solubility (scf/bbl)	25	25	25

Volumes, Rates and Qualities are based on Downhole Temperature and Pressures.

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ACID TREATMENT SCHEDULE

NITROGEN GAS SYSTEM

RATE SCHEDULE

Stage	Downhole System			Wellhead Rates			
	Clean Volume (gal)	Base Fluid	Total Rate (bpm)	Clean System (bpm)	Clean Fluid (bpm)	Nitrogen (scfm)	Diverting Agent (unit/min)
1	5000	28% HCL	12.0	12.9	4.2	14912	
2	1200	X-linked Gel	12.0	12.9	4.2	14912	
3	5000	28% HCL	12.0	12.9	4.2	14912	
4	1200	X-linked Gel	12.0	12.9	4.2	14912	
5	5000	28% HCL	12.0	12.9	4.2	14912	
6	1200	X-linked Gel	12.0	12.9	4.2	14912	
7	5000	28% HCL	12.0	12.9	4.2	14912	
8	1260	15# Gel	8.0	8.0	8.0	0	

FLUID & GAS QUANTITIES

Stage	Surface Stage Totals		Surface Cumulative Totals		Mitchell Quality			Average Specific Gravity
	Clean (bbls)	N2 (Mscf)	Clean (bbls)	N2 (Mscf)	W %	P %	F %	
1	41.7	147.94	41.7	147.94	68	64	65	0.613
2	10.0	35.51	51.7	183.45	67	64	65	0.573
3	41.7	147.94	93.3	331.39	68	64	65	0.613
4	10.0	35.51	103.3	366.89	67	64	65	0.573
5	41.7	147.94	145.0	514.84	68	64	65	0.613
6	10.0	35.51	155.0	550.34	67	64	65	0.573
7	41.7	147.94	196.7	698.28	68	64	65	0.613
8	30.0	0.00	226.7	698.28	0	0	0	0.999

NOTE: The Mitchell Quality is the Gas Rate divided by the Gas + Gel Rate.

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ACID TREATMENT SCHEDULE

NITROGEN GAS SYSTEM

PROCEDURE

Stage	Downhole Volumes			Diverting Agent					Surface Treating Pressure (psi)
	Clean			Conc. (pda)	Type	Stage (volume)	Cum (lbs)	Cum (b.s.)	
	System	Fluid							
	(gals)	(gals)	(bbls)						
1	5000	1750	41.7		Acid				5386
2	1200	420	10.0		Diverter				5450
3	5000	1750	41.7		Acid				5386
4	1200	420	10.0		Diverter				5450
5	5000	1750	41.7		Acid				5386
6	1200	420	10.0		Diverter				5450
7	5000	1750	41.7		Acid				5386
8	1260	1260	30.0		Flush				4278
Totals	24860	9520	226.7						

TREATMENT SCHEDULE

Stage	Wellhead Rates				Clean Volume Without N2		Stage Pump Time	Total Pump Time
	Clean Fluid (bpm)	N2		Diverting Agent (unit/min)				
		N2 (scfm)	Conc. (scf/bbl)		(bbls)	(cum)		
1	4.2	14912	3551		41.7	41.7	00:09:55	00:09:55
2	4.2	14912	3551		10.0	51.7	00:02:22	00:12:18
3	4.2	14912	3551		41.7	93.3	00:09:55	00:22:13
4	4.2	14912	3551		10.0	103.3	00:02:22	00:24:36
5	4.2	14912	3551		41.7	145.0	00:09:55	00:34:31
6	4.2	14912	3551		10.0	155.0	00:02:22	00:36:54
7	4.2	14912	3551		41.7	196.7	00:09:55	00:46:49
8	8.0	0	0		30.0	226.7	00:03:45	00:50:34
Total Pump Time:								00:50:34

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PRICE ESTIMATE

Product Material

QTY	UNIT	PRODUCT DESCRIPTION	UNIT PRICE	GROSS AMOUNT	DISC (%)	NET AMOUNT
70	gals	Ferrotrol 300L	33.20	2,324.00	48.0	1,208.48
7583	c-scf	Nitrogen	4.05	30,711.15	42.0	17,812.47
21	gals	NE-940	82.25	1,727.25	48.0	898.17
7000	gals	HCl, 20.1 - 28%	5.45	38,150.00	48.0	19,838.00
600	c-scf	Nitrogen, Cool Down	4.05	2,430.00	42.0	1,409.40
18	gals	Enzyme G-NE	36.70	660.60	42.0	383.15
2	gals	BF-7L	43.30	86.60	42.0	50.23
1	gals	Magnacide 575	471.00	471.00	42.0	273.18
2	gals	Inflo-102	137.00	274.00	42.0	158.92
28	gals	FAW-5 (drum)	78.75	2,205.00	42.0	1,278.90
10	gals	GW-3LD Green	90.50	905.00	42.0	524.90
8	gals	XLW-32	35.70	285.60	42.0	165.65
14	gals	CI-14	92.50	1,295.00	42.0	751.10
Product Material Subtotal:				\$81,525.20		\$44,752.55

Service Charges

QTY	UNIT	PRODUCT DESCRIPTION	UNIT PRICE	GROSS AMOUNT	DISC (%)	NET AMOUNT
1	ea	Personnel Surcharge - Acid Svc	137.00	137.00	0.0	137.00
Service Charges Subtotal:				\$137.00		\$137.00

Customer will be charged for all 'SPECIAL PROPPANTS' delivered to location, whether they are pumped or not. All proppants other than standard grade frac sand are considered 'SPECIAL PROPPANTS'.

The technical data contained in this proposal is based on the best information available at the time of writing and is subject to further analysis and testing. The pricing data contained in this proposal are estimates only and may vary depending on the work

actually performed. Pricing does not include federal, state and local taxes or royalties.

This quotation is based on BJ Services Company being awarded the work on a first call basis and within thirty (30) days of the proposal date. These prices will be subject to review if the work is done after thirty (30) days from the proposal date, or on a

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PRICE ESTIMATE

Equipment

QTY	UNIT	PRODUCT DESCRIPTION	UNIT PRICE	GROSS AMOUNT	DISC (%)	NET AMOUNT
1	2hrs	Acid Pump, 5001- 7500 psi - Slurry	4,475.00	4,475.00	42.0	2,595.50
1	2hrs	Frac Eqp Min, <1.2K HHP, Init hrs, w/gas	5,700.00	5,700.00	42.0	3,306.00
1	job	Data Acquisition, Acid, Standard	1,585.00	1,585.00	42.0	919.30
140	miles	Mileage, Heavy Vehicle	6.95	973.00	42.0	564.34
140	miles	Mileage, Auto, Pick-Up or Treating Van	3.96	554.40	42.0	321.55
2	2hrs	N2 Pump, 4-8k scfm, Over 6000 psi	5,500.00	11,000.00	42.0	6,380.00
Equipment Subtotal:				\$24,287.40		\$14,086.69

Freight/Delivery Charges

QTY	UNIT	PRODUCT DESCRIPTION	UNIT PRICE	GROSS AMOUNT	DISC (%)	NET AMOUNT
8	hrs	Bulk Delivery, Trans., Over 3000 gals	156.50	1,252.00	42.0	726.16
Freight/Delivery Charges Subtotal:				\$1,252.00		\$726.16
TOTAL:				\$107,201.60		\$59,702.40

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CONDITIONS

BJ Services' performance of services and sale of materials is expressly conditioned upon the applicability of the Terms and Conditions contained in the current BJ Services Price Book. The Terms and Conditions include, among other things, an indemnity in favor of BJ Services from Customer for damage to the well bore, reservoir damage, loss of the hole, blowouts and loss of control of the well, even if caused by the negligence or other fault of BJ Services. The Terms and Conditions also limit the warranties provided by the BJ Services and the remedies to which Customer may be entitled in the event of a breach of warranty by BJ Services. For these reasons, we strongly recommend that you carefully review a copy of the Terms and Conditions. **If you do not have a copy of the BJ Services Price Book, you can view the Terms and Conditions on BJ Services Web Site, www.bjservices.com.**

By requesting that BJ Services perform the services described herein, Customer acknowledges that such Terms and Conditions are applicable to the services. Further, by requesting the services, Customer warrants that its representative on the well location or other service site will be fully authorized to acknowledge such Terms and Conditions by executing a Field Receipt or other document presented by BJ Services containing such Terms and Conditions.

In the event that Customer and BJ Services have executed a Master Services Agreement covering the work to be performed, such Master Services Agreement shall govern in place of the Terms and

Conditions. If you are interested in entering into Master Services Agreement with BJ Services, please contact us through the "Go BJ" button on the BJ Services Web Site.



PRODUCT DESCRIPTIONS

BF-7L

A liquid pH control agent used to adjust fracturing gels into the pH range of 8.5 to 10.5. This product was designed to retain its buffering capacity at high temperatures.

CI-14

A more environmentally friendly liquid acid corrosion inhibitor for use in hydrochloric acid, HCL/HF acid, and organic acid/HCL acid systems with total acid strengths of less than 28% (by weight) in well conditions where exposure temperatures are low to moderate.

Enzyme G-NE

A patented, polymer specific enzyme breaker custom formulated to degrade polymer into non-damaging components.

FAW-5 (drum)

A general purpose foaming agent designed to foam water with nitrogen.

Ferrotrol 300L

A liquid sequestering agent used to prevent the precipitation of metal oxides.

GW-3LD Green

A guar gum gelling agent, slurried in an environmentally friendly carrier. Used to prepare hydraulic fracturing fluid systems.

HCl, 20.1 - 28%

An inorganic acid (HCL) used in primary acidizing of carbonate and sandstone formations. It is the most common of oil field acids. It is typically used in concentrations from 3% to 28% (by weight). Higher concentrations are avoided due to increased volumes of inhibitors required and quantities of fines generated as the acid spends on the formation rock.

Inflo-102

nflo-102 is a amphoteric surfactant used as a stimulation fluid additive to minimize water retention in a formation. It increases load recovery and optimizes production rapidly in hydraulic fracturing and remedial well treatments. Ideal for acid matrix or water base fracturing treatments performed in low-permeability, gas sandstone reservoirs.

Magnacide 575

Magnacide 575 represents a completely new class of antimicrobial chemistry that combines superior antimicrobial activity with a relatively benign (harmless) toxicology profile. It is effective in controlling aerobic, anaerobic, and sulfate reducing bacteria.

NE-940

A nonionic non-emulsifier with excellent load recovery capabilities in some reservoirs.

XLW-32

A liquid Borate crosslinker (Boric Acid) used in Viking gel systems.

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End of Report